SWATI AGGRAWAL

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PROFESSIONAL SUMMARY

AI Engineer who ships end-to-end: from a rough idea to a deployed, debugged, production-hardened system. Built retrieval-grounded RAG pipelines and a hybrid-retrieval recommendation API, taken through prototyping, deployment, and production debugging (OOM crashes, latency, recall gaps) rather than left as notebooks; hands-on with agentic workflows (LangGraph state graphs, tool-calling) through applied coursework. Comfortable across the full stack: Python, FastAPI, Docker, cloud deployment (AWS, Render), and LLM/agent tooling (LangChain, LangGraph, Gemini/OpenAI SDKs, vector retrieval). M.Sc. Computer Science (Sapienza Università di Roma) with an IEEE publication and a benchmarking-heavy thesis, bringing an evaluation-first habit to AI system design.

SKILLS

Core Engineering: Python, FastAPI, REST APIs, Docker, Git, Linux, debugging production issues, async request handling, data structures & systems thinking

AI & Agent Engineering: LangChain, LangGraph (state graphs, tool-calling), OpenAI SDK, Gemini API, prompt engineering, agentic workflow design

Retrieval & Grounding: RAG pipelines, chunking, embeddings, hybrid retrieval (BM25 + dense), Reciprocal Rank Fusion (RRF), re-ranking, Vector Databases (Pinecone, Chroma)

Production & Reliability: Latency optimization, caching, model preloading, LangSmith tracing/observability, AWS (ECS, Fargate, CloudWatch), Render, evaluation harnesses

ML & Deep Learning: PyTorch, TensorFlow, Scikit-learn, Computer Vision, Vision Transformers

AI SYSTEMS SHIPPED

SHL Assessment Recommender — Conversational Retrieval API github.com/swatiaggrawal/shl-assessment-recommender

- Built a conversational REST API recommending SHL assessments via **hybrid BM25 + Gemini embedding retrieval with Reciprocal Rank Fusion (RRF)**, deployed on Render.
- Hardened the service in production: debugged OOM crashes, retrieval recall gaps, pydantic validation errors, and stale-cache bugs, reaching passing behavior probes at **sub-6-second latency**.

Knowledge Copilot — Grounded RAG Chatbot

github.com/swatiaggrawal/RAG-Chatbot | Live demo: rag-chatbot-new.onrender.com

- Built a four-stage RAG pipeline (ingestion → embedding → retrieval → generation) with per-answer source citations to prevent hallucination; integrated Gemini API embeddings with Pinecone.
- Re-architected the deployment after hitting free-tier RAM/CPU limits (dropped local sentence-transformers and Chroma), shipping a stable FastAPI + React app with self-indexing startup logic.

DocDelta — Document Intelligence API github.com/swatiaggrawal/DocDelta

- Built a FastAPI microservice for semantic document comparison (Sentence Transformers, cosine similarity); containerized with Docker and deployed on AWS ECS/Fargate with CloudWatch monitoring for latency/error tracking.
- Optimized for production: async request handling, model preloading to eliminate cold starts, ~840ms warm-request latency.

EDUCATION

Sapienza Università di Roma — M.Sc. Computer Science **2026**

Thesis: Evaluating Pretrained Vision Transformers Under Continual Learning — benchmarked three modeling approaches, 90.59% avg. accuracy on Split CIFAR-100.

Sharda University — B.Tech, CSE with AI and ML **2023**

Merit scholarship (Top 5–10% of cohort)

CERTIFICATIONS

LangGraph — LangChain Academy **July 2026**

- Implemented multi-step agent workflows using LangGraph state graphs and tool-calling, working through OpenAI-to-Gemini substitutions across notebooks and LangGraph Studio.
- Debugged framework/provider incompatibilities (content-block formatting, parallel tool-call validation errors) and set up LangSmith tracing for agent-run observability.

LangChain for LLM Application Development — DeepLearning.AI **July 2026**

AI Agents in LangGraph — DeepLearning.AI **July 2026**

PUBLICATION

Machine Learning for Air Pollution — IEEE Conference Publication **2023**

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